

Matt Gresham

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OBJECTIVE

To pursue a BS in Computer Science with concentrations in AI/ML and Cybersecurity, and build a career at the intersection of software development, machine learning, and security engineering.

EDUCATION

International Community School Bangkok (ICS)

Expected Graduation: May 2027

SAT: 1240

Coursework: AP Computer Science Principles (A), AP Language & Composition, AP Cybersecurity, Precalculus, Physics, Philosophy & Worldview, Honor Band

Additional: Codecademy Intro to IT, BYU United States History

TECHNICAL SKILLS

Languages: Python, Go, JavaScript / React, HTML & CSS, Bash, Rust (learning), French (intermediate)

AI / ML: Q-Learning, Deep Q-Networks, Minimax, Alpha-Beta Pruning, PyTorch, RAG pipelines, ChromaDB, Ollama

Tools: Git/GitHub, Linux, Kali Linux, Raspberry Pi, Streamlit, Plotly, Pandas, React + Vite, SQLite

Security: DVWA, SQL Injection, XSS, Brute Forcing, Prompt Injection, Ethical Hacking

PROJECTS

Villages — Full-Stack AI Community Learning Platform

React Vite FastAPI Supabase OpenRouter Vercel

- Full-stack AI learning platform with AI-matched study cohorts, real-time discussion, AI facilitation, and a Study Hub offering Socratic tutoring, essay coaching, and study planning
- React + Vite + Tailwind frontend and a FastAPI backend, both deployed live on Vercel (serverless Python), backed by Supabase (PostgreSQL + auth + realtime), at zero cost
- Wired a GitHub Actions CI gate on pull requests; runs free-tier OpenRouter LLMs to keep inference costs at zero

Pixel-APEX — Q-Learning Game AI Agent

Python Q-Learning Minimax Plotly PyTorch

- Designed a multi-module AI architecture featuring a Q-arbiter that learns which AI advisor to trust through Q-value optimization
- Implemented tabular Q-learning with softmax exploration, decaying epsilon, and a DFS minimax opponent for training
- Built a hyperparameter sweep tool with Plotly visualization for systematic performance analysis and tuning
- Currently expanding to chess using Deep Q-Networks with experience replay and target networks

The Finance Kit — Full-Stack Finance Tracker

Python Streamlit Pandas Plotly

- Full-stack financial tool featuring both a command-line interface and a multi-page Streamlit web application
- Includes currency conversion, budget tracking, tax calculations, and interactive data visualizations
- Handles session state management, field validation, and persistent data storage across user sessions

Additional Projects

- Magellan Search Engine — Go web spider + React frontend deployed on Vercel; indexes content into SQLite
- Chess Website — Interactive React + Vite chess game with full board UI and React Router navigation
- Raspberry Pi DVWA Server — custom bash setup for practicing SQL injection, XSS, and brute forcing techniques

HACKATHONS

Hack America Hackathon — Study Assistant

2026

Python ChromaDB Ollama Streamlit

- Built and submitted an AI-powered RAG study tool that ingests Markdown files, PDFs, and images into ChromaDB for conversational querying
- Supports both local inference via Ollama and cloud models (Groq, ChatGPT, Anthropic) for low-end hardware compatibility

- Shipped a `setup.sh` script for one-command installation on Linux, macOS, and Windows, plus an auto-update feature
- Dual interface: CLI for lightweight use and Streamlit Web UI with live camera upload for photographing physical notes

USAII Hackathon

2026

- Designed, built, and submitted **Villages** (featured in Projects above) under hackathon time constraints, shipping a deployed full-stack AI learning platform end to end

COMPETITIONS & ACTIVITIES

- **NOAI (Thailand National AI Olympiad)** — Actively preparing for IOAI qualification; studying classical ML, deep learning, NLP, and computer vision
- **VEX Robotics** — Bull Run Middle School (2020–22): qualified to States as an individual competitor; came within 1 point of States on a team. ICS Bangkok (2025–26): qualified for nationals with team, declined invitation
- **Chess** — Rated approximately 1700; competitive play that integrates with AI and game theory research interests
- **Honor Band** — Trumpet at university-level repertoire; Theatre Tech — Global Citizen Award (2024)
- **Boy Scouts** — Earned Programming Merit Badge and Robotics Merit Badge (2020)

COMMUNITY SERVICE

- **House of Blessings, Bangkok (2023–24)** — Taught students Scratch programming and Canva design, helping them bring their creative ideas to life
- **Saphan Siam Foundation (2025–26)** — Helped a Bangkok-based non-profit expand its reach by building their social media presence across Line, Facebook, and Instagram; created content plans and advised on platform strategy